

PLAN: Fast and Approximate Gaussian Kernel Density Visualization in Road Networks

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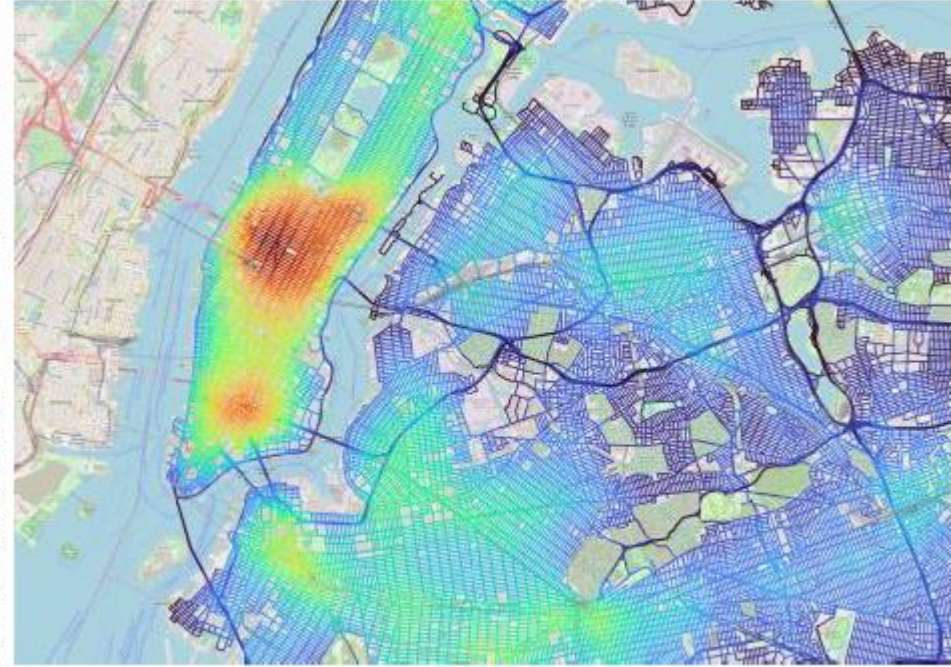
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Why Network Kernel Density Visualization (NKDV) ?



(a) New York traffic accident dataset



(b) NKDV

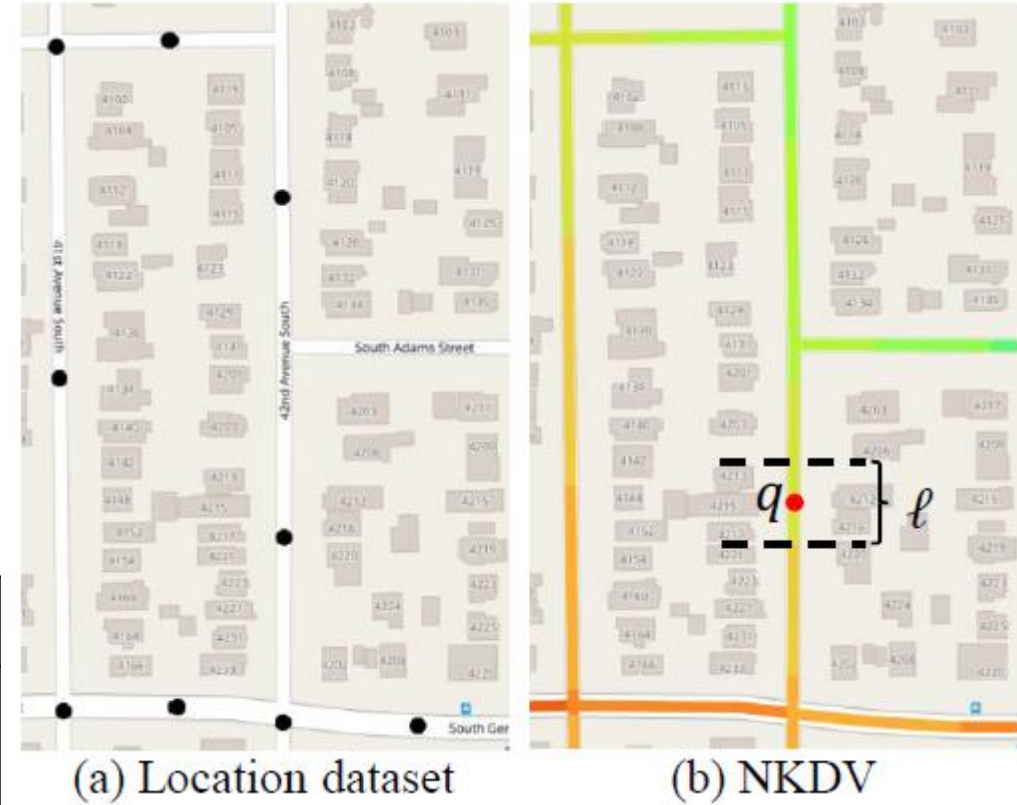
- Some representative applications:
 - Traffic/Traffic accident hotspot detection
 - Crime hotspot detection
 - Urban planning

What is NKDV?

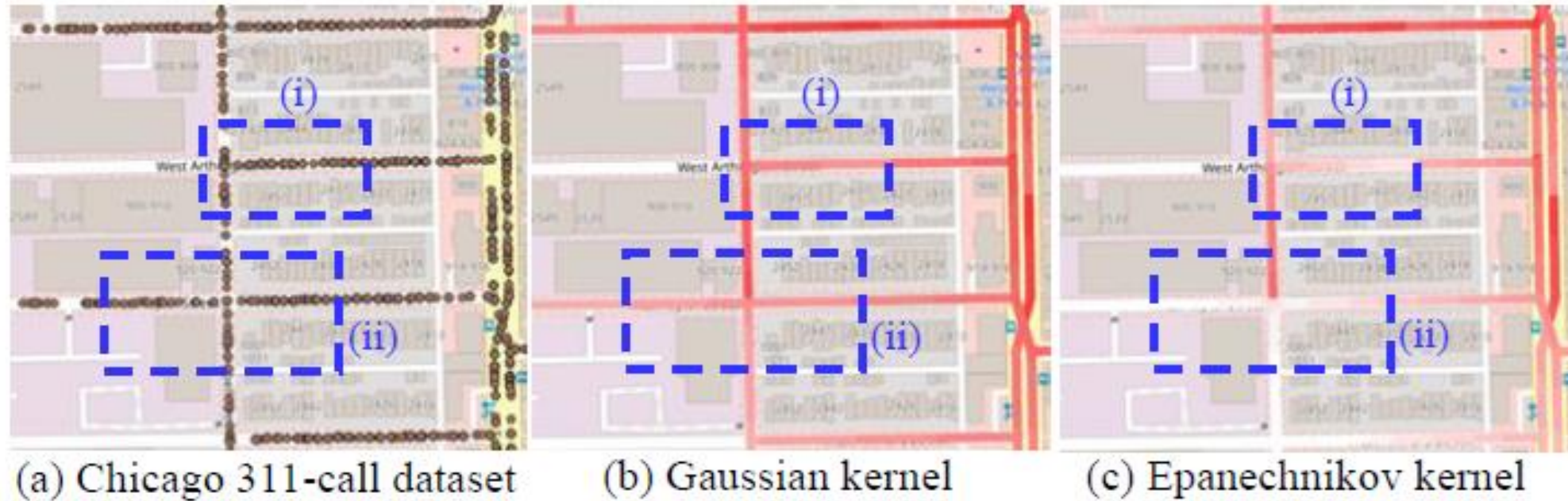
- Given a location dataset $P = \{p_1, p_2, \dots, p_n\}$ and a road network $G = (V, E)$, we need to color each lixel q based on the network kernel density function $\mathcal{F}_P(q)$.

$$\mathcal{F}_P(q) = \frac{1}{|P|} \sum_{p \in P} K_G(q, p)$$

Kernel	Function
Epanechnikov	$\begin{cases} 1 - \frac{1}{b^2} d_G(q, p)^2 & \text{if } d_G(q, p) \leq b \\ 0 & \text{otherwise} \end{cases}$
Gaussian	$\exp\left(-\frac{1}{b^2} d_G(q, p)^2\right)$



NKDV: Epanechnikov v.s. Gaussian



- Gaussian kernel can provide smoother visualization. ☺
- Gaussian kernel is the default one in many software packages. ☺

Weakness of Gaussian-based NKDV

- Gaussian-based KDV is computationally expensive. ☹️
- Example (Chicago 311-call dataset):
 - Number of data points: 10.893 million
 - Number of lixels (with 10m): 0.67 million
 - Number of operations: > 7.298 trillion
- Not feasible to generate multiple NKDVs. ☹️

Weakness of Gaussian-based NKDV

- Cannot directly extend existing fast algorithms for Gaussian-based NKDV. ☹

$$\begin{cases} 1 - \frac{1}{b^2} d_G(q, p)^2 & \text{if } d_G(q, p) \leq b \\ 0 & \text{otherwise} \end{cases} \quad \text{v.s.} \quad \exp\left(-\frac{1}{b^2} d_G(q, p)^2\right)$$

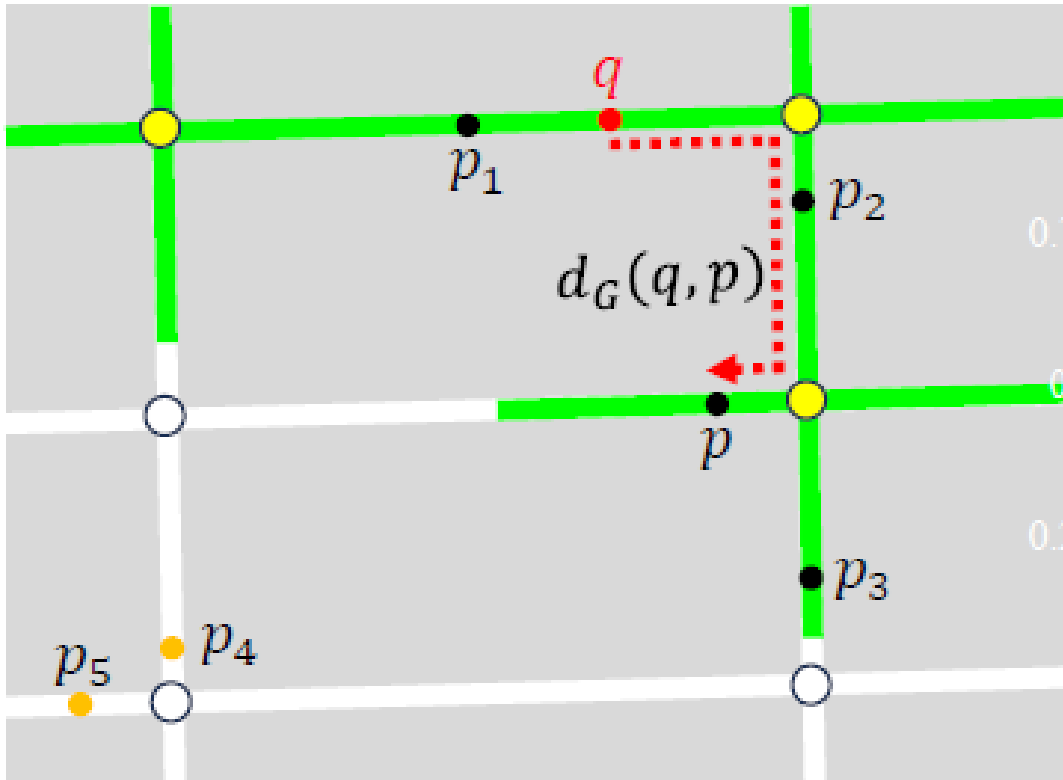
- Does not have short-tail property in Gaussian kernel.
- Cannot decompose the network kernel density function $\mathcal{F}_P(q)$.

Our Contributions

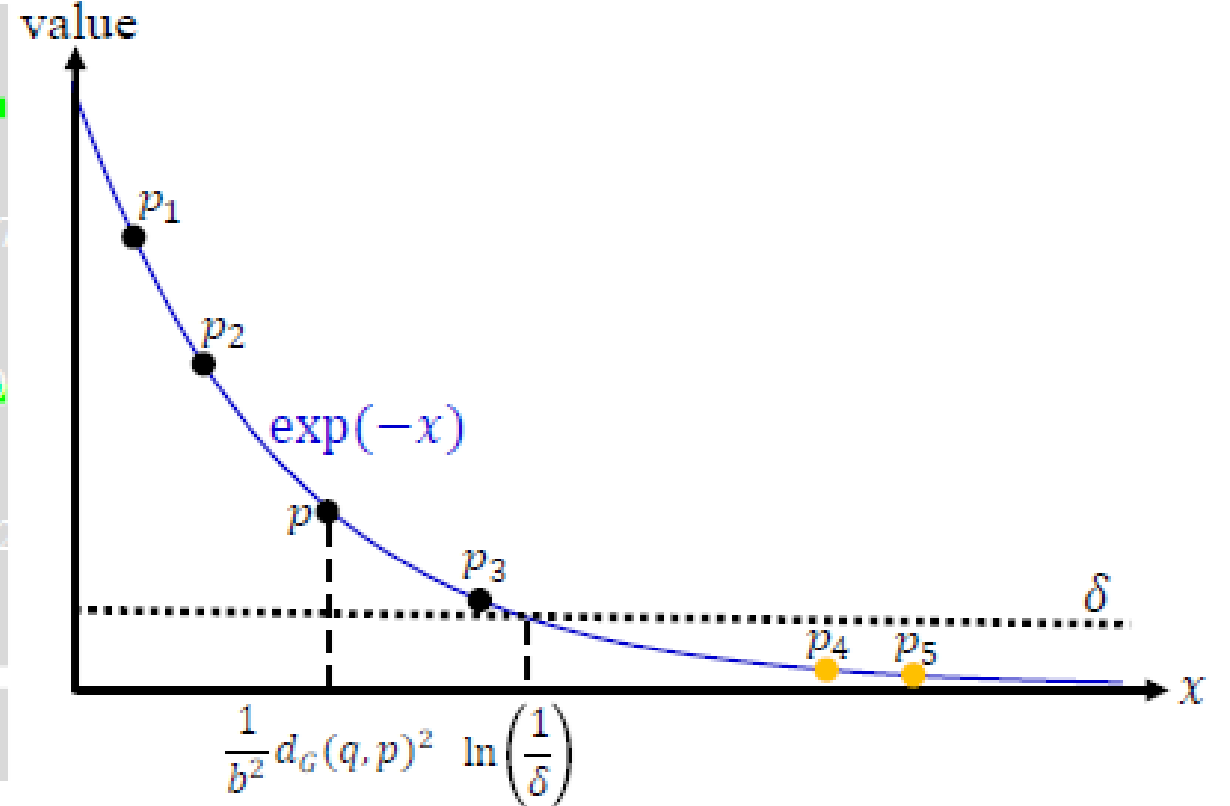
Method	Time complexity	Space complexity	Accuracy guarantee
SOTA	$O(E T_{\text{SP}} + P L)$	$O(V + P + L + S_{\text{SP}})$	Exact
PLAN	$O\left(E T_{\text{SP}} + LM E \log\left(\frac{ P }{ E }\right)\right)$ (Theorem 1)	$O(V + P + L + S_{\text{SP}} + M)$ (Theorem 3)	ϵ -absolute error (Lemma 1)
PLAN+	$O\left(E T_{\text{SP}} + LM E \log\left(\frac{ P }{M E }\right)\right)$ (Theorem 2)		

- Reduce the time complexity for generating NKDV with (1) a non-trivial approximation guarantee and (2) a slight overhead in terms of space complexity. 😊
- Achieve 32.47x to 2936.88x speedups compared with SOTA.

Core Idea 1: Enable the Short Tail Property



(a) Data points and a lixel q in a road network



(b) Exponential function

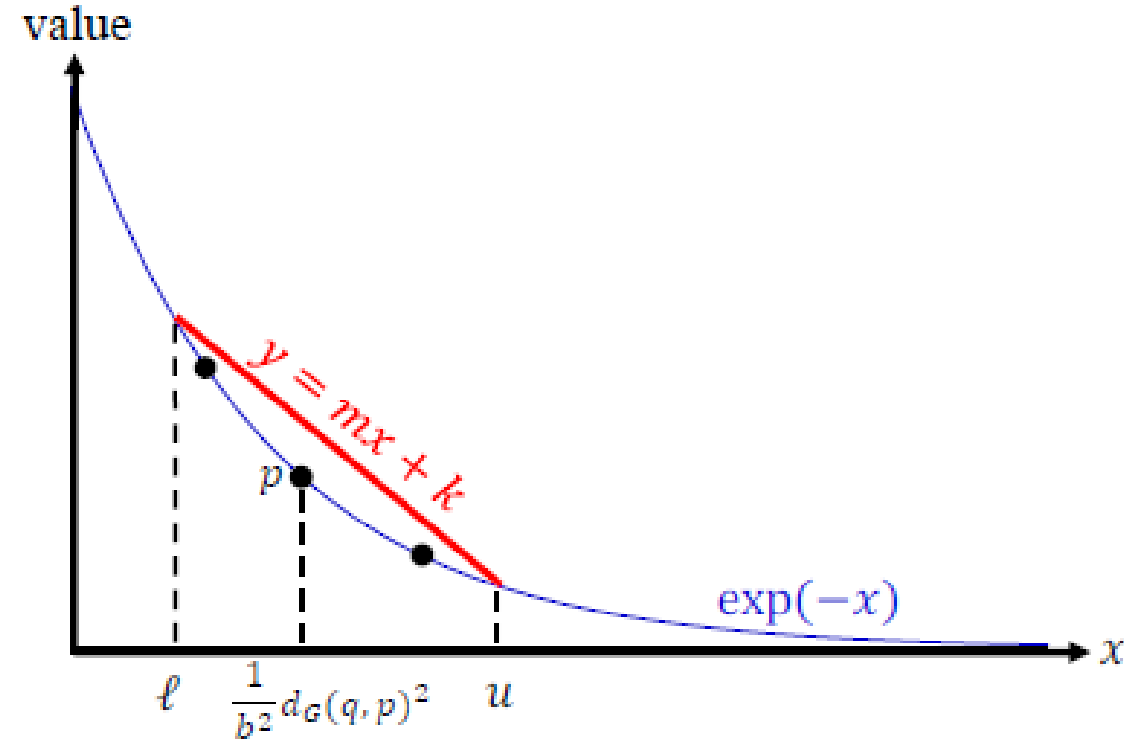
- Data points that are too far away from q can be discarded.

Core Idea 2: Function Approximation

- This upper bound function can approximate $\mathcal{F}_{\mathbb{P}}(q)$.

$$\begin{aligned}\mathcal{F}_{\mathbb{P}}(q) &= \frac{1}{|\mathbb{P}|} \sum_{p \in \mathbb{P}} \exp\left(-\frac{1}{b^2} d_G(q, p)^2\right) \\ &\leq \frac{1}{|\mathbb{P}|} \sum_{p \in \mathbb{P}} \left(\frac{m}{b^2} d_G(q, p)^2 + k\right)\end{aligned}$$

$$\text{where } \mathbb{P} = \left\{p \in P : \ell \leq \frac{1}{b^2} d_G(q, p)^2 \leq u\right\}$$



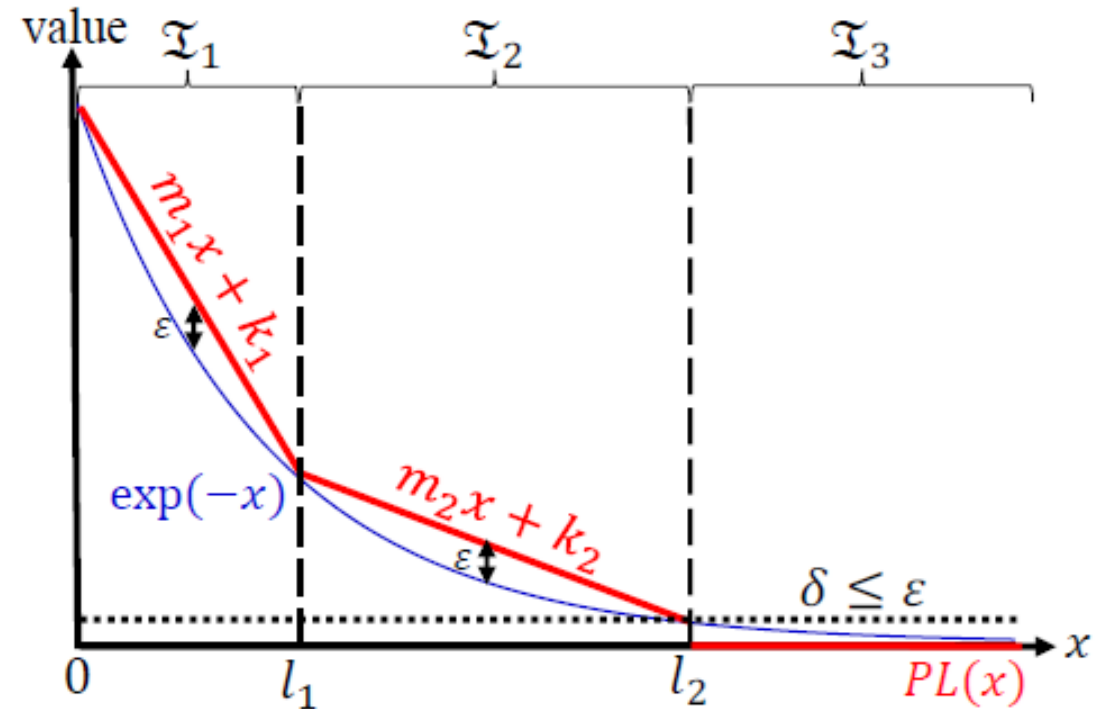
PLAN: Approximate the Network Kernel Density Function

- Use $A_P(q)$ to approximate $\mathcal{F}_P(q)$.

$$A_P(q) = \frac{1}{|P|} \sum_{p \in P} PL \left(\frac{1}{b^2} d_G(q, p)^2 \right)$$

where

$$PL(x) = \begin{cases} m_1x + k_1 & \text{if } x \in \mathcal{I}_1 \\ m_2x + k_2 & \text{if } x \in \mathcal{I}_2 \\ \vdots & \vdots \\ m_{M-1}x + k_{M-1} & \text{if } x \in \mathcal{I}_{M-1} \\ 0 & \text{if } x \in \mathcal{I}_M \end{cases}$$

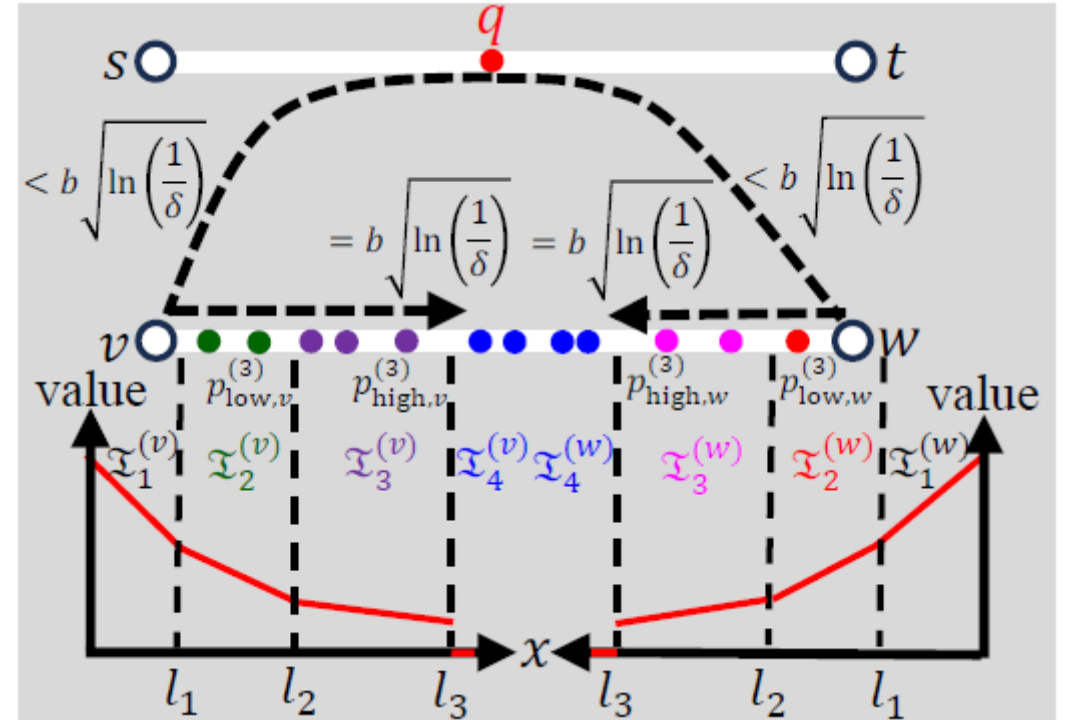


- Can achieve ε -absolute error guarantee. ☺

PLAN: Efficient Computation

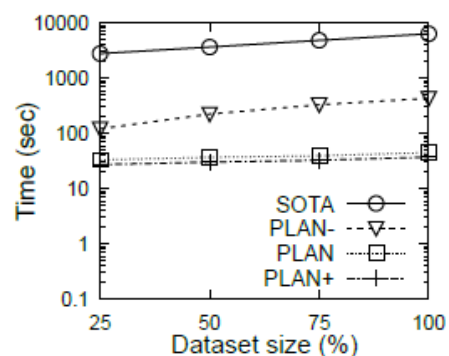
- Similar to ADA [a].
 - Perform augmentation for each data point.
 - Adopt the binary search method.
- Main difference
 - Adopt the binary search method M times.

Refer to our paper for details.

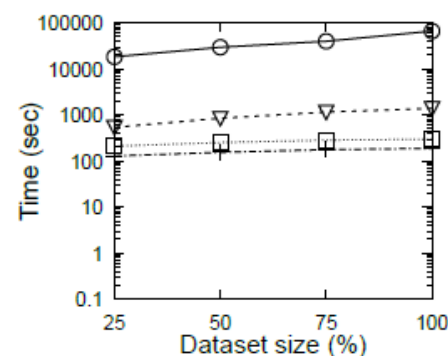


Experimental Evaluation

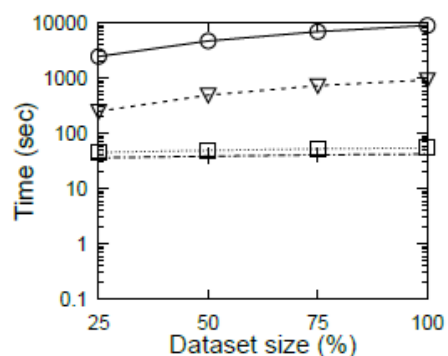
Dataset	$ V $	$ E $	$ P $	Category
Minneapolis [36]	7,388	12,913	1,598,877	911 calls
New York [37]	71,019	120,623	1,897,418	Traffic accidents
San Francisco [34]	6,627	11,593	2,985,502	Crime events
Chicago [25]	41,111	70,067	10,893,129	311 calls



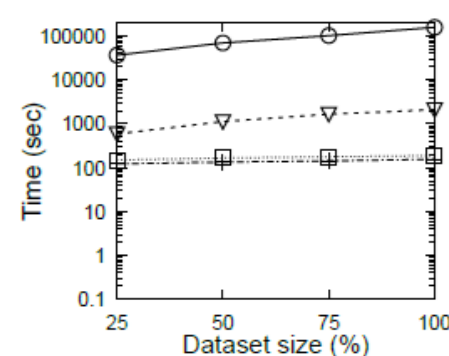
(a) Minneapolis



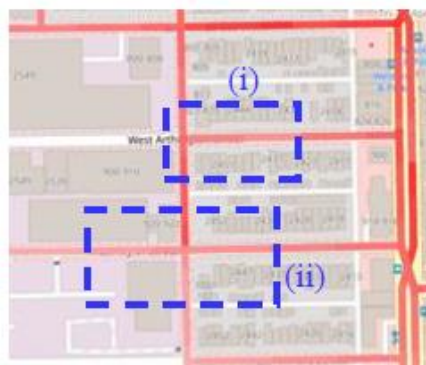
(b) New York



(c) San Francisco



(d) Chicago



(a) EXACT



(b) $\epsilon = 0.01$



(c) $\epsilon = 0.05$



(d) $\epsilon = 0.09$

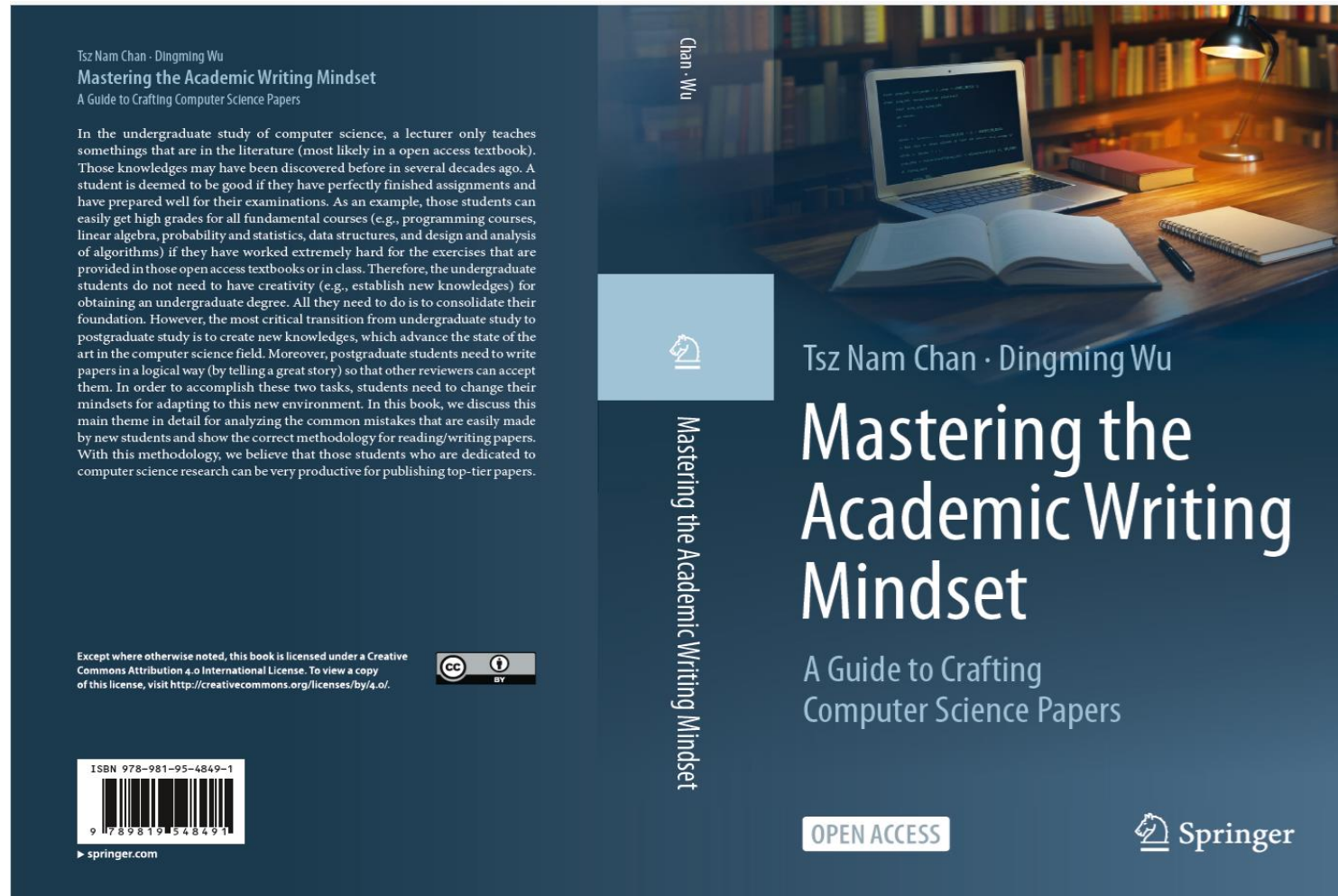


(e) Epanechnikov kernel

Future Opportunities

- Incorporate PLAN into the QGIS plugin, called Fast Density Analysis.
- Support NKDV with adaptive bandwidth.
- Adopt parallel/distributed/hardware-based approaches to improve the efficiency of NKDV.
- Support other inefficient road network analysis tasks, e.g., network Moran's I, network K-function, and network DBSCAN.

Our New Book



<https://link.springer.com/book/10.1007/978-981-95-4850-7>